

(Please write your Exam Roll No.)

Exam Roll No: _____

END TERM EXAMINATION**THIRD SEMESTER [B.TECH] DECEMBER 2025 - JANUARY 2026**
Paper Code: CIC-209 • Subject: Data Structures

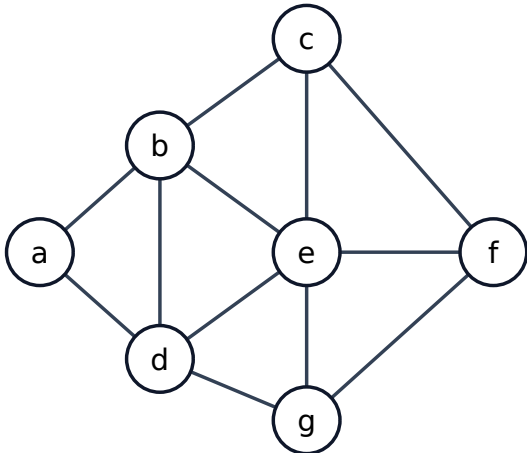
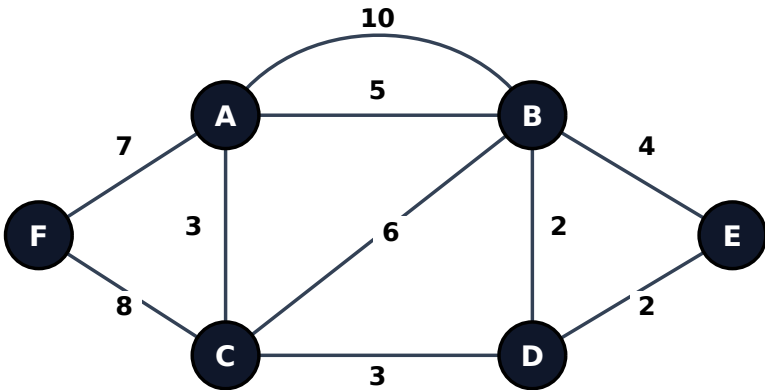
Time: 3 Hours

Maximum Marks: 60

Note: Attempt all questions as directed. Internal choice is indicated.

Q. No.	Question	Max. Marks	CO(s)
Q1	Attempt any five of the following questions:	(4×5=20)	CO1-4
1 (a)	What is data structure? Explain different classification of data structures.	[4]	CO1
1 (b)	Write a function/algorithm to implement two stacks using single array.	[4]	CO2
1 (c)	What is sparse matrix? Explain its types and memory representations.	[4]	CO1
1 (d)	The pre-order and in-order traversal of a binary tree are given below. Draw corresponding binary tree. Write a recursive algorithm for post-order traversal. Write its equivalent post-order traversal. Pre-order: A, B, D, E, G, H, C, I, F In-order: D, B, G, E, H, A, I, C, F	[4]	CO3
1 (e)	Explain the following External sorting with the help of example: (i) Natural merge, (ii) Balanced merge, and (iii) Polyphase merge.	[4]	CO4
1 (f)	What is hashing? Explain any 4 hash functions with example.	[4]	CO4
1 (g)	Consider the following specification of the directed graph G: $V(G) = \{A, B, C, D, E\}$ $E(G) = \{(A, B), (A, C), (B, C), (C, D), (E, A), (E, B)\}$ Draw its Adjacency matrix, Incidence matrix and Adjacency list.	[4]	CO3
1 (h)	Explain Union Find Algorithm.	[4]	CO4
UNIT - I	Answer Question 2 OR Question 3	[10]	CO2
Q2 (a)	Write an algorithm to convert infix expression to prefix expression using stack. Convert the following infix expression to prefix using a stack: (A - B) / C * ((D ^ E * F)	[5]	CO2
Q2 (b)	What is a DEQUE? What is its type? Write an algorithm/function for deletion from the rear end of a DEQUE.	[5]	CO2
OR	— Alternative Choice for Unit I —		
Q3 (a)	What is an array? What are the advantages and disadvantages of an array? How is a multidimensional array represented in memory?	[5]	CO1
Q3 (b)	Write a function/algorithm to reverse a singly linked list.	[5]	CO2
UNIT - II	Answer Question 4 OR Question 5	[10]	CO3
Q4 (a)	Construct an AVL search tree by inserting the following elements in the order of their occurrence: 74, 1, 34, 16, 3, 100, 90	[5]	CO3
Q4 (b)	Explain the insertion of a new value in the max binary heap with the help of example.	[5]	CO3
OR	— Alternative Choice for Unit II —		
Q5 (a)	What is Binary Search tree? Construct a Binary Search tree for the following data in sequence: A, D, E, L, M, K, C, P, Z, Q, X	[5]	CO3
Q5 (b)	What is B-tree? What are the properties of B-tree? Construct B-tree of order 3 by inserting the following elements in the order of their occurrence: 110, 120, 130, 140, 150, 15, 255, 200, 600	[5]	CO3
UNIT - III	Answer Question 6 OR Question 7	[10]	CO4
Q6 (a)	Write an algorithm/function for Quicksort.	[5]	CO4
Q6 (b)	Given list of numbers: 110, 120, 124, 133, 145, 167, 178, 199, 210, 220, 230. Search the position of 220 using Binary Search algorithm.	[5]	CO4

[P.T.O.]

Q. No.	Question	Max. Marks	CO(s)
OR	— Alternative Choice for Unit III (Continued from Page 1) —		
Q7 (a)	What is collision resolution? Explain different techniques of collision resolution with the help of example.	[5]	CO4
Q7 (b)	Write an algorithm/function for Insertion sort. Perform insertion sort on the following values: 34, 54, 24, 84, 4, 64, 74, 14	[5]	CO4
UNIT - IV	Answer Question 8 OR Question 9	[10]	CO3/4
Q8 (a)	Traverse the following graph using Breadth First Search (BFS) algorithm: 	[5]	CO3
Q8 (b)	Explain the following terms used in graph with the help of example: (i) Isolated vertex (ii) Transitive closure (iii) Minimum Spanning Tree (iv) Complete graph (v) Connected graph	[5]	CO3
OR	— Alternative Choice for Unit IV —		
Q9 (a)	What is Minimum Cost Spanning Tree? Generate MST for the following graph using Kruskal's Algorithm: 	[5]	CO4
Q9 (b)	Explain Dijkstra's Algorithm with the help of example.	[5]	CO4

**** End of Question Paper ****